

Mathematical Expectations

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The mathematical expectation of a discrete r.v. X with pmf $p(x)$ is given as

$$E(X) = \sum_x x p(x)$$

$$\text{provided } \sum_x |x| p(x) < \infty$$

The mathematical expectation of a continuous r.v. X with pdf $f(x)$ is given as

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

$$\text{provided } \int_{-\infty}^{\infty} |x| f(x) dx < \infty$$

Expected value of function of a random variable

Consider a r.v. X with pdf (pmf) $f(x)$ $p(x)$ and distribution function $F_X(x)$. If $g(\cdot)$ is a function such that $g(X)$ is a random variable and $E[g(X)]$ exists (i.e. is defined), then

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f(x) dx \quad (\text{if } X \text{ is Cont. r.v.})$$

$$E[g(X)] = \sum_x g(x) p(x) \quad (\text{if } X \text{ is discrete r.v.})$$

Remark: Extension for higher dimension.

If X and Y have a joint pdf $f(x, y)$ and

$$Z = h(X, Y) \quad \text{then } \infty$$
$$E(Z) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x, y) f(x, y) dx dy \quad (\text{if } X \text{ and } Y \text{ are Cont. r.v.})$$

$$E(Z) = \sum_x \sum_y h(x, y) p(x, y) \quad (\text{if } X \text{ and } Y \text{ are discrete r.v.})$$

Moments in terms of expectation.

$$\mu'_r = E(X^r) = \int_{-\infty}^{\infty} x^r f(x) dx \quad \text{if } X \text{ is Cont. r.v.}$$

$$= \sum_x x^r p(x) \quad \text{if } X \text{ is discrete r.v.}$$

$$\mu'_1 = E(X) = \text{Mean}, \quad \mu'_2 = E(X^2) \quad (\text{second moment about origin})$$

Central Moments

$$\begin{aligned} \mu_r &= E(X - E(X))^r = \int_{-\infty}^{\infty} (x - \mu_1')^r f(x) dx \quad \text{for Cont. r.v.} \\ &= \sum_x (x - \mu_1')^r p(x) \quad \text{for discrete r.v.} \end{aligned}$$

$$\begin{aligned} \mu_2 &= E(X - E(X))^2 = E(X - \bar{x})^2 = \int_{-\infty}^{\infty} (x - \bar{x})^2 f(x) dx \quad \text{for Cont. r.v.} \\ &= \sum_x (x - \bar{x})^2 p(x) \quad \text{for discrete r.v.} \end{aligned}$$

Similarly we can find μ_3, μ_4 etc.

Expectation of a Constant and its effect over R.V.

For a Constant C

$$E(C) = C$$

$$E(CX) = C E(X)$$

$$E(A + BX) = A + B E(X) \quad ; \quad A + B \text{ are some constants.}$$

Properties of Expectation

i) Addition Rule.

$$E(X_1 + X_2 + \dots + X_n) = E(X_1) + E(X_2) + \dots + E(X_n)$$

$$\Rightarrow E\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n E(X_i)$$

ii) Multiplication Rule

If $X_1, X_2, \dots, X_n$ are independent random variables

$$E(X_1 X_2 \dots X_n) = E(X_1) E(X_2) \dots E(X_n)$$

$$\Rightarrow E\left(\prod_{i=1}^n X_i\right) = \prod_{i=1}^n E(X_i)$$

iii) $E(a_1 X_1 + a_2 X_2 + \dots + a_n X_n) = a_1 E(X_1) + a_2 E(X_2) + \dots + a_n E(X_n)$

$$\Rightarrow E\left(\sum_{i=1}^n a_i X_i\right) = \sum_{i=1}^n a_i E(X_i)$$

iv) $E\left(\frac{1}{X}\right) \neq \frac{1}{E(X)} \quad ; \quad E(X^{\frac{1}{2}}) \neq (E(X))^{\frac{1}{2}}$

$$E[\log X] \neq \log E(X) \quad E(X^2) \neq (E(X))^2$$

v) If $X \geq 0$ $E(X) \geq 0$

vi) If $Y \leq X$
 $E(Y) \leq E(X)$

vii) $|E(X)| \leq E|X|$ provided the expectation exists.

Properties of Variance

$$V(a + bX) = b^2 V(X)$$

Proof. Let $Y = a + bX$

$$E(Y) = a + bE(X)$$

$$Y - E(Y) = b[X - E(X)]$$

$$V(Y) = E(Y - E(Y))^2 = b^2 E(X - E(X))^2$$

$$\Rightarrow \underline{V(Y) = V(a + bX) = b^2 V(X)}$$

Corr. If $a = 0$, $V(bX) = b^2 V(X)$

if $b = 1$ $V(a + X) = V(X)$

if $b = 0$, $V(a) = 0$, variance of a constant is zero.

Covariance: The Covariance between X and Y is defined as

$$\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))]$$

$$= E(XY) - E(X)E(Y)$$

If X and Y are independent.

$$\text{Cov}(X, Y) = E(X)E(Y) - E(X)E(Y) = 0$$

Remark i) $\text{Cov}(aX, bY) = ab \text{Cov}(X, Y)$

ii) $\text{Cov}(X + a, Y + b) = \text{Cov}(X, Y)$

iii) $\text{Cov}\left(\frac{X - \bar{X}}{\sigma_X}, \frac{Y - \bar{Y}}{\sigma_Y}\right) = \frac{1}{\sigma_X \sigma_Y} \text{Cov}(X, Y)$

iv) $\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y)$

v) $\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$

$$vi) \text{Cov}(ax+by, cx+dy) = ac\sigma_x^2 + bd\sigma_y^2 + (ad+bc)\text{Cov}(x,y)$$

$$vii) \text{ If } x \text{ and } y \text{ indep. } \text{Cov}(x,y) = 0$$

Variance of a Linear Combination of R.V.s.

If $x_1, x_2, \dots, x_n$ are r.v.s.

$$V\left(\sum_{i=1}^n a_i x_i\right) = \sum_{i=1}^n a_i^2 V(x_i) + 2 \sum_{i=1}^n \sum_{j=1, j \neq i}^n a_i a_j \text{Cov}(x_i, x_j)$$

$$\text{or } \sum_{i=1}^n a_i^2 V(x_i) + \sum_{i=1}^n \sum_{j=1, j \neq i}^n a_i a_j \text{Cov}(x_i, x_j)$$

$$\underline{\text{Ex.}} \quad V(x_1 \pm x_2) = V(x_1) + V(x_2) \pm 2\text{Cov}(x_1, x_2)$$

$$\text{If } x_1 \text{ and } x_2 \text{ are independent. } \text{Cov}(x_1, x_2) = 0$$

$$V(x_1 \pm x_2) = V(x_1) \pm V(x_2)$$

Ex.

x	-3	6	9
$P(X=x) = p(x)$	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{1}{3}$

Find $E(X)$, $E(X^2)$ and $E(2X+1)^2$

$$E(X) = \sum_x x p(x) = -3 \times \frac{1}{6} + 6 \times \frac{1}{2} + 9 \times \frac{1}{3} = \frac{11}{2}$$

$$E(X^2) = \sum_x x^2 p(x) = 9 \times \frac{1}{6} + 36 \times \frac{1}{2} + 81 \times \frac{1}{3} = \frac{93}{2}$$

$$\begin{aligned} E(2X+1)^2 &= E(4X^2 + 1 + 4X) \\ &= 4E(X^2) + 1 + 4E(X) \\ &= 4 \times \frac{93}{2} + 1 + 4 \times \frac{11}{2} = 209 \end{aligned}$$

Ex. (a) Find the expectation of the number on a die when thrown.

* (b) Can be generalised for n fair dice
 $S = \sum_{i=1}^n X_i$
 $E(S) = \sum_{i=1}^n E(X_i)$
 $= \frac{7}{2} \cdot n$

(b) Two unbiased dice are thrown. Find the expected values of the sum of numbers of points on them.

(a) Let X be the r.v. representing the number on a die when thrown. X can take any one of the values 1, 2, 3, 4, 5, 6 with equal prob $\frac{1}{6}$. Hence $E(X) = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + \dots + 6 \cdot \frac{1}{6} = \frac{1}{6} \cdot \frac{6 \times 7}{2} = \frac{7}{2}$

* (b) X_i be the number obtained on the i th dice ($i=1, 2$)
 $S = X_1 + X_2$ $E(S) = E(X_1) + E(X_2) = \frac{7}{2} + \frac{7}{2} = 7$

Characteristic Function

In some cases m.g.f. does not exist, since the ~~the~~ integral $\int_{-\infty}^{\infty} e^{tx} f(x) dx$ or the series $\sum_x e^{tx} p(x)$ does not converge absolutely for real values of t for some distributions.

Defⁿ: The characteristic function of a r.v. X is defined as

$$\begin{aligned}\phi_X(t) &= E e^{itx} = \int_{-\infty}^{\infty} e^{itx} f(x) dx && \text{for cont. r.v.} \\ &= \sum_x e^{itx} p(x) && \text{for discrete r.v.}\end{aligned}$$

Obviously $\phi_X(t)$ is a complex valued function of real variable t . it may be noted that

$$\begin{aligned}|\phi(t)| &= \left| \int e^{itx} f(x) dx \right| \leq \int |e^{itx}| f(x) dx \\ &= \int f(x) dx = 1\end{aligned}$$

$$\begin{aligned}\text{Since } e^{itx} &= |\cos tx + i \sin tx|^{\frac{1}{2}} = (\cos^2 tx + \sin^2 tx)^{\frac{1}{2}} \\ &= 1.\end{aligned}$$

Since $|\phi(t)| \leq 1$ characteristic function $\phi_X(t)$ always exists.

Properties of characteristic Functions

For all real t , we have

$$(i) \quad \phi(0) = \int_{-\infty}^{\infty} f(x) dx = 1$$

$$(ii) \quad |\phi(t)| \leq 1 = \phi(0)$$

$$(iii) \quad \phi(t) \text{ is continuous function of } t \text{ in } (-\infty, \infty)$$

Result 1 If X is a r.v. with characteristic function $\phi_X(t)$ and if $\mu_r' = E(X^r)$ exists, then

$$\mu_r' = (-i)^r \left[\frac{\partial^r \phi_X(t)}{\partial t^r} \right]_{t=0}.$$

Proof: $\phi_X(t) = E e^{itX} = \int_{-\infty}^{\infty} e^{itx} f(x) dx$

$$\frac{\partial \phi_X(t)}{\partial t} = \int_{-\infty}^{\infty} e^{itx} f(x) \cdot (ix) dx$$

$$\begin{aligned} \left. \frac{\partial \phi_X(t)}{\partial t} \right|_{t=0} &= \int_{-\infty}^{\infty} e^0 f(x) \cdot ix dx \\ &= i \mu_1' \Rightarrow \mu_1' = \left(\frac{1}{i} \right) \left[\frac{\partial \phi_X(t)}{\partial t} \right]_{t=0} \end{aligned}$$

$$\left. \frac{\partial^2 \phi_X(t)}{\partial t^2} \right|_{t=0} = \int_{-\infty}^{\infty} e^0 f(x) (ix)^2 dx$$

$$\Rightarrow \left. \frac{\partial^2 \phi_X(t)}{\partial t^2} \right|_{t=0} = i^2 \mu_2' \Rightarrow \mu_2' = \left(\frac{1}{i} \right)^2 \left[\frac{\partial^2 \phi_X(t)}{\partial t^2} \right]_{t=0}$$

In general, we get $\mu_r' = \left(\frac{1}{i} \right)^r \left[\frac{\partial^r \phi_X(t)}{\partial t^r} \right]_{t=0}$

$$\Rightarrow \boxed{\mu_r' = (-i)^r \left[\frac{\partial^r \phi_X(t)}{\partial t^r} \right]_{t=0}}$$

Result 2 $\phi_{cX}(t) = \phi_X(ct)$ c , being a constant.

Result 3 : If $X_1, X_2, \dots, X_n$ are independent r.v.s.

$$\phi_{X_1+X_2+\dots+X_n}(t) = \phi_{X_1}(t) \phi_{X_2}(t) \dots \phi_{X_n}(t)$$

Converse is not true for $n=2$.

i.e. if $\phi_{X_1+X_2}(t) = \phi_{X_1}(t) \phi_{X_2}(t)$ does not imply X_1, X_2 are independent.

Result 4 : Effect of change of origin + scale on Page 7
Characteristic Function.

If $U = \frac{x-a}{h}$, a + h being constants.

$$\begin{aligned}\phi_U(t) &= E e^{iUt} = E e^{i\left(\frac{x-a}{h}\right)t} \\ &= e^{-\frac{iat}{h}} E e^{i\frac{x}{h}t}\end{aligned}$$

$$\boxed{\phi_U(t) = e^{-\frac{iat}{h}} \phi_X\left(\frac{t}{h}\right)}$$

